

SIDON DIFFERENCE AVOIDANCE

ABSTRACT. We prove that, for every fixed integer $M \geq 1$ and all sufficiently large N , every non-empty Sidon set $A \subseteq \{1, \dots, N\}$ has a Sidon difference-avoiding companion $B \subseteq \{1, \dots, N\}$ of size M , namely $(A - A) \cap (B - B) = \{0\}$. The main ingredient is a finite Fourier compactness theorem for dense independent sets in Cayley graphs whose forbidden-difference indicator has almost non-negative Fourier transform. As a consequence, the first question in Erdős Problem 43 has a negative answer. We also give a self-contained Bose–Chowla parity construction showing that the equal-cardinality strengthening in Erdős Problem 43 is false in the strongest asymptotic sense allowed by the classical upper bound for Sidon sets.

1. INTRODUCTION

Write $[N] = \{1, \dots, N\}$. A finite set A of integers is a *Sidon set* if

$$a_1 + a_2 = a_3 + a_4, \quad a_i \in A,$$

implies $\{a_1, a_2\} = \{a_3, a_4\}$. Equivalently, each non-zero ordered difference $a - a'$ is represented by at most one ordered pair $(a, a') \in A^2$.

Erdős Problem 42 asks whether, for every fixed M , every Sidon set $A \subseteq [N]$ has a Sidon set $B \subseteq [N]$ of size M with no common non-zero difference, once N is sufficiently large in terms of M [2]. The equality $(A - A) \cap (B - B) = \{0\}$ is impossible when $A = \emptyset$, so throughout the affirmative statement is made for non-empty A ; for $A = \emptyset$ the natural avoidance conclusion is simply $(A - A) \cap (B - B) \subseteq \{0\}$.

Our first main result proves this problem. The underlying principle is that, if A is Sidon, then 1_{A-A} is a small negative perturbation of a positive-definite function. A compactness argument shows that a dense set cannot fail to contain many finite configurations avoiding any such forbidden-difference set.

Theorem 1.1 (Erdős Problem 42). *For every integer $M \geq 1$ there is $N_0(M)$ such that the following holds for all $N \geq N_0(M)$. If $A \subseteq [N]$ is a non-empty Sidon set, then there is a Sidon set $B \subseteq [N]$ with $|B| = M$ and*

$$(A - A) \cap (B - B) = \{0\}.$$

If $A = \emptyset$, the same proof gives the corresponding inclusion $(A - A) \cap (B - B) \subseteq \{0\}$.

Let $f(N)$ denote the maximum cardinality of a Sidon subset of $[N]$. Erdős Problem 43 asks whether pairs of Sidon sets $A, B \subseteq [N]$ satisfying $(A - A) \cap (B - B) = \{0\}$ must obey

$$\binom{|A|}{2} + \binom{|B|}{2} \leq \binom{f(N)}{2} + O(1),$$

and whether, under the additional hypothesis $|A| = |B|$, this can be improved to

$$\binom{|A|}{2} + \binom{|B|}{2} \leq (1 - c + o(1)) \binom{f(N)}{2}$$

for some absolute $c > 0$ [4]. Tao observed in the discussion of Problem 42 that a positive solution to Problem 42 gives a negative answer to the first question in Problem 43 by taking A to have size $f(N)$ [3]. We record the resulting formal consequence.

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Corollary 1.2 (First part of Erdős Problem 43). *There is no absolute constant C such that*

$$\binom{|A|}{2} + \binom{|B|}{2} \leq \binom{f(N)}{2} + C$$

for every N and every pair of Sidon sets $A, B \subseteq [N]$ satisfying $(A - A) \cap (B - B) = \{0\}$. More quantitatively, for every $C \geq 0$ and all sufficiently large N depending on C , such pairs exist with

$$\binom{|A|}{2} + \binom{|B|}{2} > \binom{f(N)}{2} + C.$$

For the equal-cardinality assertion in Problem 43, the obstruction is different. A Bose–Chowla modular Sidon set has all non-zero directed differences distinct; splitting it into even and odd residues nearly bisects it, since the two within-parity directed-difference systems must pack into the non-zero even residues. This gives two equal-size Sidon sets with disjoint difference sets and with essentially maximal total difference count.

Theorem 1.3 (Second part of Erdős Problem 43). *For infinitely many N there are Sidon sets $A, B \subseteq [N]$ such that*

$$|A| = |B|, \quad (A - A) \cap (B - B) = \{0\},$$

and

$$\binom{|A|}{2} + \binom{|B|}{2} \geq (1 - o(1)) \binom{f(N)}{2}.$$

Consequently there is no absolute constant $c > 0$ for which the equal-cardinality bound in Erdős Problem 43 holds for all sufficiently large N .

The proof of Theorem 1.1 occupies Sections 2 and 3. The proof of Theorem 1.3 is given in Section 4.

2. A FINITE FOURIER AVOIDANCE THEOREM

For a prime p write $G_p = \mathbb{Z}/p\mathbb{Z}$. For $h : G_p \rightarrow \mathbb{C}$ define

$$\widehat{h}(\xi) = \mathbb{E}_{x \in G_p} h(x) e^{-2\pi i \xi x / p}, \quad \xi \in G_p.$$

Then

$$h(x) = \sum_{\xi \in G_p} \widehat{h}(\xi) e^{2\pi i \xi x / p}, \quad \sum_{\xi \in G_p} |\widehat{h}(\xi)|^2 = \mathbb{E}_{x \in G_p} |h(x)|^2.$$

For a compact abelian group Z with Haar probability measure μ , its character group $\widehat{Z}$ is written additively and characters are written multiplicatively. Thus $(\gamma + \gamma')(z) = \gamma(z)\gamma'(z)$ and $(-\gamma)(z) = \overline{\gamma(z)}$. For $h \in L^1(Z)$ we use

$$\widehat{h}(\gamma) = \int_Z h(z) \overline{\gamma(z)} d\mu(z).$$

Theorem 2.1 (Finite Fourier avoidance). *Fix $m \geq 1$, $\alpha > 0$ and $\rho > 0$. There are constants*

$$\varepsilon = \varepsilon(m, \alpha, \rho) > 0, \quad c = c(m, \alpha, \rho) > 0,$$

and a prime threshold $p_0 = p_0(m, \alpha, \rho)$ such that the following holds for every prime $p \geq p_0$. Let $F, U \subseteq G_p$ satisfy

$$F = -F, \quad 0 \in F, \quad |F| \leq (1 - \rho)p, \quad |U| \geq \alpha p,$$

and

$$\widehat{\mathbf{1}_F}(\xi) \geq -\varepsilon \quad (\xi \in G_p).$$

Then

$$\#\{(x_1, \dots, x_m) \in U^m : x_i - x_j \notin F \text{ for all } i \neq j\} \geq cp^m.$$

Proof. Suppose the theorem is false. Then there are primes $p_n \rightarrow \infty$, sets $F_n, U_n \subseteq G_n := \mathbb{Z}/p_n\mathbb{Z}$, and $\varepsilon_n \downarrow 0$ such that

$$F_n = -F_n, \quad 0 \in F_n, \quad |F_n| \leq (1 - \rho)p_n, \quad |U_n| \geq \alpha p_n, \\ \widehat{\mathbf{1}_{F_n}}(\xi) \geq -\varepsilon_n \quad (\xi \in G_n),$$

but the normalized number of desired m -tuples tends to zero. Put

$$f_n = \mathbf{1}_{F_n}, \quad u_n = \mathbf{1}_{U_n}.$$

Since $F_n = -F_n$, this means

$$(1) \quad \Lambda_n := \mathbb{E}_{x_1, \dots, x_m \in G_n} \prod_{i=1}^m u_n(x_i) \prod_{1 \leq i < j \leq m} (1 - f_n(x_i - x_j)) \longrightarrow 0.$$

After passing to a subsequence, assume also that

$$(2) \quad \mathbb{E}_{G_n} u_n \longrightarrow \beta \geq \alpha.$$

The limiting compact group. Set $\eta_r = 2^{-r}$ and

$$S_{n,r} = \{\xi \in G_n : |\widehat{f_n}(\xi)| > \eta_r\}.$$

By Parseval, $|S_{n,r}| \leq \eta_r^{-2}$. Passing to a diagonal subsequence, we may label these spectra compatibly by finite sets

$$A_1 \subseteq A_2 \subseteq \dots$$

so that, for each fixed r and all sufficiently large n ,

$$S_{n,r} = \{a_n : a \in A_r\},$$

with $a \mapsto a_n$ injective on A_r . Let $A_* = \bigcup_r A_r$, and let $\mathbb{Z}^{(A_*)}$ be the free abelian group on A_* . If $q \in \mathbb{Z}^{(A_*)}$, let $q_n \in G_n$ be the corresponding integral linear combination of the labelled frequencies. Since $\mathbb{Z}^{(A_*)}$ is countable, another diagonal passage ensures that, for every $q \in \mathbb{Z}^{(A_*)}$, either $q_n = 0$ for all sufficiently large n , or $q_n \neq 0$ for all sufficiently large n .

Define

$$R = \{q \in \mathbb{Z}^{(A_*)} : q_n = 0 \text{ for all sufficiently large } n\}, \quad \Gamma = \mathbb{Z}^{(A_*)}/R.$$

For $\gamma \in \Gamma$, choose a representative q and write $\gamma_n = q_n$ for all sufficiently large n ; this is well-defined up to changing finitely many n . For every finite $C \subseteq \Gamma$, the map $\gamma \mapsto \gamma_n$ is injective on C for all sufficiently large n , because if $\gamma \neq \gamma'$, then $\gamma - \gamma' \notin R$.

The group Γ is torsion-free. Indeed, if $k\gamma = 0$ in Γ and $k \neq 0$, then $k\gamma_n = 0$ eventually. Since $p_n > |k|$ eventually, multiplication by k is invertible on G_n , so $\gamma_n = 0$ eventually and hence $\gamma = 0$.

Let

$$Z = \widehat{\Gamma}.$$

By the standard Pontryagin duality criterion, Z is compact and connected. Passing to one further diagonal subsequence, assume that for every $\gamma \in \Gamma$ the limits

$$a_\gamma = \lim_{n \rightarrow \infty} \widehat{f_n}(\gamma_n), \quad b_\gamma = \lim_{n \rightarrow \infty} \widehat{u_n}(\gamma_n)$$

exist. Since $F_n = -F_n$, the coefficients $\widehat{f_n}(\xi)$ are real; the lower Fourier bound gives

$$(3) \quad a_\gamma \geq 0 \quad (\gamma \in \Gamma).$$

The limiting functions. Let

$$P(z) = \sum_{\gamma \in C} c_\gamma \overline{\gamma(z)}$$

be a trigonometric polynomial on Z , with $C \subseteq \Gamma$ finite. For all sufficiently large n define

$$P_n(x) = \sum_{\gamma \in C} c_\gamma e^{-2\pi i \gamma_n x / p_n}, \quad x \in G_n.$$

If $P \geq 0$ on Z , then $P_n \geq 0$ on G_n for all sufficiently large n . To see this, note that the subgroup of relations among the finite set C is finitely generated; all those relations lie in R and hence hold among the corresponding γ_n for all sufficiently large n . Thus, for each $x \in G_n$, the map $\gamma \mapsto e^{2\pi i \gamma_n x / p_n}$ defines a character on the subgroup generated by C , and every character on a subgroup of a discrete abelian group extends to the whole group. Hence the values of P_n are values of P .

Also,

$$\mathbb{E}_{G_n} P_n \longrightarrow \int_Z P d\mu,$$

because every non-trivial $\gamma \in \Gamma$ has $\gamma_n \neq 0$ eventually. Define linear functionals on trigonometric polynomials by

$$L_f(P) = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} f_n P_n, \quad L_u(P) = \lim_{n \rightarrow \infty} \mathbb{E}_{G_n} u_n P_n.$$

The limits exist by the definitions of a_γ, b_γ . If $P \geq 0$, then

$$0 \leq L_f(P) \leq \int_Z P d\mu, \quad 0 \leq L_u(P) \leq \int_Z P d\mu.$$

The trigonometric polynomials are uniformly dense in $C(Z)$. Positivity implies boundedness of L_f, L_u , so they extend uniquely to positive linear functionals on $C(Z)$. The domination by Haar measure extends from non-negative trigonometric polynomials to non-negative continuous functions by uniform approximation: if $\phi \geq 0$ and Q is real-valued with $\|\phi - Q\|_\infty < \delta$, then $Q + \delta \geq 0$ and

$$L_f(\phi) \leq L_f(Q) + \delta L_f(1) \leq L_f(Q + \delta) \leq \int_Z (Q + \delta) d\mu \leq \int_Z \phi d\mu + 2\delta,$$

and similarly for L_u .

By the Riesz representation theorem and Radon–Nikodym theorem there are measurable functions $f, u : Z \rightarrow [0, 1]$ such that

$$L_f(P) = \int_Z f P d\mu, \quad L_u(P) = \int_Z u P d\mu$$

for every trigonometric polynomial P . Taking $P = \bar{\gamma}$ gives

$$(4) \quad \hat{f}(\gamma) = a_\gamma, \quad \hat{u}(\gamma) = b_\gamma \quad (\gamma \in \Gamma).$$

In particular,

$$(5) \quad \int_Z f d\mu \leq 1 - \rho, \quad \int_Z u d\mu = \beta \geq \alpha,$$

and

$$(6) \quad \hat{f}(\gamma) \geq 0 \quad (\gamma \in \Gamma).$$

Convergence of pattern counts. For $E \subseteq \binom{[m]}{2}$, define

$$T_{E,n}(h, v) = \mathbb{E}_{x_1, \dots, x_m \in G_n} \prod_{i=1}^m v(x_i) \prod_{ij \in E} h(x_i - x_j),$$

where ij always means $i < j$. For functions on Z define

$$T_E(h, v) = \int_{Z^m} \prod_{i=1}^m v(z_i) \prod_{ij \in E} h(z_i - z_j) d\mu(z_1) \cdots d\mu(z_m).$$

We claim that

$$(7) \quad T_{E,n}(f_n, u_n) \longrightarrow T_E(f, u) \quad \left(E \subseteq \binom{[m]}{2} \right).$$

Choose finite sets $B_r \subseteq \Gamma$ such that

$$A_r \subseteq B_r, \quad B_r = -B_r, \quad 0 \in B_r, \quad B_r \subseteq B_{r+1}, \quad \bigcup_r B_r = \Gamma.$$

Since every finite subset of Γ generates a finitely generated torsion-free abelian group, there is a finite non-empty set $Q_r \subseteq \Gamma$ such that

$$(8) \quad \frac{|Q_r \cap (Q_r - \gamma)|}{|Q_r|} \geq 1 - \eta_r \quad (\gamma \in B_r).$$

Define

$$K_{n,r}(x) = \frac{1}{|Q_r|} \left| \sum_{\gamma \in Q_r} e^{2\pi i \gamma_n x / p_n} \right|^2, \quad x \in G_n.$$

For fixed r and all sufficiently large n , the map $\gamma \mapsto \gamma_n$ is injective on $Q_r - Q_r$. Hence $K_{n,r} \geq 0$, $\mathbb{E}_{G_n} K_{n,r} = 1$, and all Fourier coefficients of $K_{n,r}$ lie in $[0, 1]$. More precisely, for $\delta \in Q_r - Q_r$,

$$(9) \quad \widehat{K}_{n,r}(\delta_n) = \frac{|Q_r \cap (Q_r - \delta)|}{|Q_r|}.$$

Let

$$g_{n,r} = f_n * K_{n,r}, \quad (a * b)(x) = \mathbb{E}_{y \in G_n} a(x - y)b(y).$$

Then $0 \leq g_{n,r} \leq 1$ and $\widehat{g}_{n,r} = \widehat{f}_n \widehat{K}_{n,r}$. For fixed r and all sufficiently large n ,

$$(10) \quad \sup_{\xi \in G_n} |\widehat{f}_n(\xi) - \widehat{g}_{n,r}(\xi)| \leq \eta_r.$$

Indeed, if $|\widehat{f}_n(\xi)| > \eta_r$, then $\xi = \gamma_n$ for some $\gamma \in A_r \subseteq B_r$, and (8)–(9) give $1 - \widehat{K}_{n,r}(\xi) \leq \eta_r$. If $|\widehat{f}_n(\xi)| \leq \eta_r$, then (10) follows from $0 \leq \widehat{K}_{n,r}(\xi) \leq 1$.

For $h : G_n \rightarrow \mathbb{C}$ define the Cayley cut norm

$$\|h\|_{\square,n} = \sup_{|a|,|b| \leq 1} |\mathbb{E}_{x,y \in G_n} h(x - y)a(x)b(y)|.$$

Fourier inversion and Cauchy–Schwarz give

$$(11) \quad \|h\|_{\square,n} \leq \sup_{\xi \in G_n} |\widehat{h}(\xi)|.$$

Therefore

$$(12) \quad \|f_n - g_{n,r}\|_{\square,n} \leq \eta_r.$$

A one-edge telescoping argument gives

$$(13) \quad |T_{E,n}(f_n, u_n) - T_{E,n}(g_{n,r}, u_n)| \leq |E| \eta_r.$$

When a single edge ij is replaced by $f_n - g_{n,r}$ and all other variables are fixed, all remaining factors involving x_i and x_j are bounded by 1 and factor as a function of x_i times a function of x_j , so (12) applies.

For fixed r , put $D_r = Q_r - Q_r$. Expanding the finite Fourier series for $g_{n,r}$ in $T_{E,n}(g_{n,r}, u_n)$, every term is indexed by a map $\omega : E \rightarrow D_r$. Define

$$\sigma_i(\omega) = \sum_{j: i < j, ij \in E} \omega(ij) - \sum_{j: j < i, ji \in E} \omega(ji).$$

The corresponding contribution is

$$\prod_{ij \in E} \widehat{f}_n(\omega(ij)_n) \widehat{K}_{n,r}(\omega(ij)_n) \prod_{i=1}^m \widehat{u}_n(-\sigma_i(\omega)_n).$$

The finite set consisting of D_r and all sums $\sigma_i(\omega)$ is fixed once r and E are fixed, so the construction of Γ gives all coefficient limits. Hence

$$(14) \quad \lim_{n \rightarrow \infty} T_{E,n}(g_{n,r}, u_n) = T_E(g_r, u),$$

where

$$K_r(z) = \frac{1}{|Q_r|} \left| \sum_{\gamma \in Q_r} \gamma(z) \right|^2, \quad g_r = f * K_r.$$

Here convolution on Z is with respect to Haar probability measure.

The kernels K_r are a positive approximate identity. They satisfy $K_r \geq 0$ and $\int K_r d\mu = 1$; for every fixed $\gamma \in \Gamma$, (8) and the exhaustion $B_r \nearrow \Gamma$ imply $\widehat{K_r}(\gamma) \rightarrow 1$. Thus the probability measures $K_r d\mu$ converge weakly to the point mass at 0. By L^1 -continuity of translations,

$$(15) \quad \|g_r - f\|_1 = \|f * K_r - f\|_1 \rightarrow 0.$$

Since $0 \leq f, g_r, u \leq 1$, another telescoping argument gives

$$(16) \quad T_E(g_r, u) \rightarrow T_E(f, u).$$

Combining (13), (14), and (16), and then letting $r \rightarrow \infty$, proves (7).

By inclusion-exclusion and (7),

$$(17) \quad \Lambda_n \rightarrow \Lambda := \int_{Z^m} \prod_{i=1}^m u(z_i) \prod_{1 \leq i < j \leq m} (1 - f(z_i - z_j)) d\mu^m.$$

Together with (1), this gives $\Lambda = 0$.

The positive-definite endpoint. We shall show that $\Lambda > 0$, obtaining the contradiction. Recall that

$$0 \leq f \leq 1, \quad \int_Z f d\mu \leq 1 - \rho, \quad \widehat{f}(\gamma) \geq 0 \quad (\gamma \in \Gamma).$$

First we prove that f has an absolutely convergent Fourier series. Let $C \subseteq \Gamma$ be finite and $\tau > 0$. Choose a finite non-empty $Q \subseteq \Gamma$ such that

$$\frac{|Q \cap (Q - \gamma)|}{|Q|} \geq 1 - \tau \quad (\gamma \in C \cup (-C)).$$

This is possible because $C \cup (-C)$ generates a finitely generated free abelian group. Set

$$K_Q(z) = \frac{1}{|Q|} \left| \sum_{\eta \in Q} \eta(z) \right|^2.$$

Then $K_Q \geq 0$, $\int K_Q d\mu = 1$, and

$$\widehat{K_Q}(\gamma) = \frac{|Q \cap (Q - \gamma)|}{|Q|}.$$

Since $0 \leq f \leq 1$ and all $\widehat{f}(\gamma)$ are non-negative,

$$1 \geq \int_Z f K_Q d\mu = \sum_{\gamma \in \Gamma} \widehat{f}(\gamma) \widehat{K_Q}(-\gamma) \geq (1 - \tau) \sum_{\gamma \in C} \widehat{f}(\gamma).$$

Letting $\tau \downarrow 0$ gives

$$\sum_{\gamma \in C} \widehat{f}(\gamma) \leq 1.$$

As C was arbitrary,

$$(18) \quad \sum_{\gamma \in \Gamma} \widehat{f}(\gamma) \leq 1.$$

Thus

$$F(z) = \sum_{\gamma \in \Gamma} \widehat{f}(\gamma) \gamma(z)$$

converges absolutely and uniformly. It has the same Fourier coefficients as f , so $F = f$ almost everywhere. We replace f by this continuous representative; the bounds $0 \leq f \leq 1$ then hold everywhere, since Haar measure has full support.

Let

$$H = \{z \in Z : f(z) = 1\}.$$

We claim that

$$(19) \quad \mu(H) = 0.$$

If $H = \emptyset$ there is nothing to prove. Otherwise choose $h_0 \in H$. Let

$$v = (\sqrt{\widehat{f}(\gamma)})_{\gamma \in \Gamma} \in \ell^2(\Gamma),$$

which is legitimate by (18), and let U_z be the unitary representation of Z on $\ell^2(\Gamma)$ given by $U_z e_\gamma = \gamma(z) e_\gamma$. With the inner product linear in the first variable,

$$f(z) = \langle U_z v, v \rangle.$$

Since $f(h_0) = 1$,

$$1 = |\langle U_{h_0} v, v \rangle| \leq \|U_{h_0} v\| \|v\| = \|v\|^2 = f(0) \leq 1.$$

Thus equality holds throughout, $\|v\| = 1$, and $U_{h_0} v = v$. The same argument shows that, for every $h \in Z$,

$$f(h) = 1 \iff U_h v = v.$$

Therefore H is the stabilizer of v , hence a closed subgroup of Z .

A closed subgroup of a compact group with positive Haar measure is open, for instance by Steinhaus' theorem. Since Z is connected, it has no proper open subgroup. If $\mu(H) > 0$, then $H = Z$, so $f \equiv 1$, contradicting $\int f d\mu \leq 1 - \rho$. This proves (19).

Now set $S = \{z \in Z : u(z) > 0\}$. By (5), $\mu(S) > 0$. For each pair $i < j$, Fubini and (19) give

$$\mu^m \{(z_1, \dots, z_m) \in S^m : z_i - z_j \in H\} = 0.$$

Thus, for almost every $(z_1, \dots, z_m) \in S^m$, one has $z_i - z_j \notin H$ for all $i < j$. On this positive-measure set,

$$\prod_{i=1}^m u(z_i) > 0, \quad \prod_{1 \leq i < j \leq m} (1 - f(z_i - z_j)) > 0.$$

Therefore $\Lambda > 0$, contradicting (17). The theorem follows. $\square$

3. PROOF OF PROBLEM 42 AND THE FIRST PART OF PROBLEM 43

Proof of Theorem 1.1. Fix $M \geq 1$. Apply Theorem 2.1 with

$$m = M, \quad \alpha = \frac{1}{8}, \quad \rho = \frac{1}{2},$$

and let the resulting constants be ε_0, c_0, p_0 .

Let N be sufficiently large in terms of M , and let $A \subseteq [N]$ be a non-empty Sidon set. Put $k = |A|$. By Bertrand's postulate, choose a prime

$$4N < p < 8N.$$

Embed $[N]$ into G_p . Since $p > 2N$, all differences between elements of $[N]$ are represented modulo p without wraparound.

Set $F = A - A \subseteq G_p$. Then $F = -F$ and $0 \in F$. Since A is Sidon, every non-zero ordered difference $a - a'$ is represented at most once. Hence

$$|F| = 1 + k(k-1).$$

The positive differences of A are distinct and lie in $\{1, \dots, N-1\}$, so

$$\binom{k}{2} \leq N-1.$$

Consequently

$$(20) \quad |F| = 1 + k(k-1) \leq 2N-1 < \frac{p}{2}.$$

We next verify the Fourier lower bound. For $t \in G_p$ let

$$r(t) = \#\{(a, a') \in A^2 : a - a' = t\}.$$

Then $r(0) = k$, while $r(t) = \mathbf{1}_F(t)$ for $t \neq 0$. Thus

$$\mathbf{1}_F = r - (k-1)\mathbf{1}_{\{0\}}.$$

For every $\xi \in G_p$,

$$\widehat{r}(\xi) = \frac{1}{p} \left| \sum_{a \in A} e^{-2\pi i \xi a/p} \right|^2 \geq 0,$$

and hence

$$(21) \quad \widehat{\mathbf{1}}_F(\xi) \geq -\frac{k-1}{p}.$$

Since $\binom{k}{2} \leq N-1$ and $p > 4N$, the right-hand side of (21) is $o_{N \rightarrow \infty}(1)$. Taking N large enough in terms of M , we have

$$\widehat{\mathbf{1}}_F(\xi) \geq -\varepsilon_0 \quad (\xi \in G_p), \quad p \geq p_0.$$

Let $U = [N] \subseteq G_p$. Since $p < 8N$, $|U| = N > p/8$. Theorem 2.1, together with (20), gives at least $c_0 p^M$ ordered tuples

$$(x_1, \dots, x_M) \in U^M$$

such that

$$(22) \quad x_i - x_j \notin F \quad (i \neq j).$$

Since $0 \in F$, every such tuple has distinct entries.

We discard tuples whose non-zero ordered differences are not all distinct. For two distinct ordered pairs $(i, j) \neq (r, s)$ with $i \neq j$ and $r \neq s$, the relation

$$x_i - x_j = x_r - x_s$$

is a non-trivial linear equation in $x_1, \dots, x_M$, and has $O_M(N^{M-1})$ solutions in U^M . There are $O_M(1)$ choices of the two ordered pairs, so the total number of bad tuples is $O_M(N^{M-1})$. For N still larger, this is less than $c_0 p^M$.

Choose a tuple satisfying (22) whose non-zero ordered differences are all distinct, and set

$$B = \{x_1, \dots, x_M\} \subseteq [N].$$

Then $|B| = M$, and the distinctness of non-zero ordered differences says exactly that B is Sidon. If a non-zero integer difference of B belonged to $A - A$, then its residue modulo p would lie in F , contradicting (22). Hence

$$(A - A) \cap (B - B) \subseteq \{0\}.$$

Since A and B are non-empty, both difference sets contain 0, and equality follows. $\square$

Proof of Corollary 1.2. Let $C \geq 0$. Choose an integer M with $\binom{M}{2} > C$. For all sufficiently large N , let $A \subseteq [N]$ be a Sidon set of maximum size $f(N)$. By Theorem 1.1, there is a Sidon set $B \subseteq [N]$ with $|B| = M$ and $(A - A) \cap (B - B) = \{0\}$. Therefore

$$\binom{|A|}{2} + \binom{|B|}{2} = \binom{f(N)}{2} + \binom{M}{2} > \binom{f(N)}{2} + C.$$

This rules out any absolute $O(1)$ term. $\square$

4. BALANCED COUNTEREXAMPLES FOR THE SECOND PART OF PROBLEM 43

We need only the upper half of the classical asymptotic formula $f(N) = (1 + o(1))\sqrt{N}$.

Lemma 4.1 (Erdős–Turán upper estimate). *One has*

$$f(N) \leq \sqrt{N} + O(N^{1/4}).$$

Consequently,

$$\binom{f(N)}{2} \leq \frac{N}{2} + O(N^{3/4}).$$

Proof. Let $A \subseteq [N]$ be Sidon, and put $k = |A|$. For $H \geq 1$ let $I = [H]$ and consider $g = \mathbf{1}_A * \mathbf{1}_I$ on the integers. Since A is Sidon, the ordered difference representation function

$$r_A(d) = \#\{(a, a') \in A^2 : a - a' = d\}$$

satisfies $r_A(0) = k$ and $r_A(d) \leq 1$ for $d \neq 0$. Hence

$$\begin{aligned} \|g\|_2^2 &= \sum_d r_A(d) (\mathbf{1}_I * \mathbf{1}_{-I})(d) \\ &\leq kH + \sum_{0 < |d| < H} (H - |d|) \leq kH + H^2. \end{aligned}$$

On the other hand, $\|g\|_1 = kH$ and g is supported on an interval of length at most $N + H$, so Cauchy's inequality gives

$$\frac{k^2 H^2}{N + H} \leq kH + H^2.$$

Taking $H = \lceil N^{3/4} \rceil$ and using the trivial bound $k = O(\sqrt{N})$, which follows from $\binom{k}{2} \leq N - 1$, yields

$$k^2 \leq N + O(N^{3/4}).$$

Thus $k \leq \sqrt{N} + O(N^{1/4})$, uniformly in A . $\square$

A subset $S \subseteq \mathbb{Z}/m\mathbb{Z}$ is called Sidon modulo m if $s_1 + s_2 = s_3 + s_4$ in $\mathbb{Z}/m\mathbb{Z}$ implies $\{s_1, s_2\} = \{s_3, s_4\}$.

Theorem 4.2 (Bose–Chowla, order two). *For every prime power q there is a set*

$$S \subseteq \{1, 2, \dots, q^2 - 2\}$$

of cardinality q which is Sidon modulo $q^2 - 1$.

Proof. Let θ be a generator of $\mathbb{F}_{q^2}^\times$. Define $S \subseteq \mathbb{Z}/(q^2 - 1)\mathbb{Z}$ by

$$S = \{a : \theta^a - \theta \in \mathbb{F}_q\}.$$

For each $x \in \mathbb{F}_q$, the element $\theta + x$ is non-zero and hence equals θ^a for a unique residue class a modulo $q^2 - 1$. Thus $|S| = q$. Also $0 \notin S$, since $1 - \theta \notin \mathbb{F}_q$, so S has representatives in $\{1, \dots, q^2 - 2\}$.

Suppose $a, b, c, d \in S$ and $a + b \equiv c + d \pmod{q^2 - 1}$. Choose $x, y, z, w \in \mathbb{F}_q$ such that

$$\theta^a = \theta + x, \quad \theta^b = \theta + y, \quad \theta^c = \theta + z, \quad \theta^d = \theta + w.$$

Then

$$(\theta + x)(\theta + y) = (\theta + z)(\theta + w).$$

Since $1, \theta$ form a basis of $\mathbb{F}_{q^2}$ over $\mathbb{F}_q$, comparison of coefficients gives

$$x + y = z + w, \quad xy = zw.$$

Therefore $\{x, y\} = \{z, w\}$, and hence $\{a, b\} = \{c, d\}$. So S is Sidon modulo $q^2 - 1$. $\square$

Lemma 4.3. *Let S be a Sidon set in an abelian group G . If $d \neq 0$ and*

$$s_1 - s_2 = d = s_3 - s_4, \quad s_i \in S,$$

then $(s_1, s_2) = (s_3, s_4)$.

Proof. The equality $s_1 - s_2 = s_3 - s_4$ gives $s_1 + s_4 = s_3 + s_2$. By the Sidon property, $\{s_1, s_4\} = \{s_3, s_2\}$. The matching $s_1 = s_2$ and $s_4 = s_3$ would force $d = 0$, so the only possible matching is $s_1 = s_3$ and $s_4 = s_2$. $\square$

Proposition 4.4. *Let q be an odd prime power, and set*

$$N = \frac{q^2 - 1}{2}.$$

Then there are Sidon sets $A, B \subseteq [N]$ such that $(A - A) \cap (B - B) = \{0\}$,

$$|A| = |B| \geq \frac{q}{2} - O(\sqrt{q}),$$

and hence

$$\binom{|A|}{2} + \binom{|B|}{2} \geq \frac{N}{2} - O(N^{3/4}).$$

Proof. Let $M = q^2 - 1 = 2N$, and let

$$S \subseteq \{1, 2, \dots, M - 1\}$$

be a Bose–Chowla Sidon set modulo M with $|S| = q$. Split S according to the parity of its integer representatives:

$$S_{\text{even}} = S \cap 2\mathbb{Z}, \quad S_{\text{odd}} = S \cap (2\mathbb{Z} + 1),$$

and write $a = |S_{\text{even}}|$, $b = |S_{\text{odd}}|$, so $a + b = q$.

We claim that

$$(23) \quad a(a - 1) + b(b - 1) \leq N - 1.$$

Indeed, in $\mathbb{Z}/M\mathbb{Z}$, the directed differences between two distinct elements of S_{even} are $a(a - 1)$ distinct non-zero even residues, by Lemma 4.3. The same holds for the $b(b - 1)$ directed differences inside S_{odd} . These two collections are disjoint, again by Lemma 4.3, because an equal non-zero difference would force the corresponding ordered pairs in S to be identical, which is impossible across the parity classes. There are exactly $N - 1$ non-zero even residues modulo M , proving (23).

Since

$$a(a - 1) + b(b - 1) = \frac{q^2 + (a - b)^2}{2} - q,$$

comparison with (23) gives

$$(a - b)^2 \leq 2q - 3.$$

Thus

$$\min(a, b) \geq \frac{q - \sqrt{2q - 3}}{2}.$$

Let $m = \min(a, b)$, and choose subsets

$$E \subseteq S_{\text{even}}, \quad O \subseteq S_{\text{odd}}, \quad |E| = |O| = m.$$

Define

$$A_0 = \{e/2 : e \in E\} \subseteq \{0, \dots, N - 1\}, \quad B_0 = \{(o - 1)/2 : o \in O\} \subseteq \{0, \dots, N - 1\},$$

and set $A = A_0 + 1$, $B = B_0 + 1$. Then $A, B \subseteq [N]$ and $|A| = |B| = m$.

We prove that A_0 is Sidon; the proof for B_0 is identical. If

$$x_1 + x_2 = x_3 + x_4, \quad x_i \in A_0,$$

write $x_i = e_i/2$ with $e_i \in E$. Then $e_1 + e_2 = e_3 + e_4$ in the integers, hence modulo M . Since S is Sidon modulo M , $\{e_1, e_2\} = \{e_3, e_4\}$, and so $\{x_1, x_2\} = \{x_3, x_4\}$.

Finally, suppose $d \in (A_0 - A_0) \cap (B_0 - B_0)$. Then for some $e_1, e_2 \in E$ and $o_1, o_2 \in O$,

$$d = \frac{e_1 - e_2}{2} = \frac{o_1 - o_2}{2}.$$

Thus $e_1 - e_2 = o_1 - o_2$ as integers and hence modulo M . If this common residue is non-zero, Lemma 4.3 forces $(e_1, e_2) = (o_1, o_2)$, impossible by parity. Therefore the common residue is zero; since all representatives lie in $\{1, \dots, M - 1\}$, this gives $e_1 = e_2$ and $o_1 = o_2$, so $d = 0$. Hence $(A_0 - A_0) \cap (B_0 - B_0) = \{0\}$, and the same holds after translating to A, B .

The lower bound for the total difference count follows from

$$\binom{|A|}{2} + \binom{|B|}{2} = m(m-1) \geq \left(\frac{q}{2} - O(\sqrt{q})\right) \left(\frac{q}{2} - O(\sqrt{q}) - 1\right) = \frac{q^2}{4} - O(q^{3/2}),$$

and $q^2 = 2N + 1$. □

Proof of Theorem 1.3. Take $N = (q^2 - 1)/2$ along any sequence of odd prime powers $q \rightarrow \infty$, for instance odd primes. Proposition 4.4 gives Sidon sets $A, B \subseteq [N]$ with equal cardinalities, disjoint non-zero difference sets, and

$$\binom{|A|}{2} + \binom{|B|}{2} \geq \frac{N}{2} - O(N^{3/4}).$$

By Lemma 4.1,

$$\binom{f(N)}{2} \leq \frac{N}{2} + O(N^{3/4}).$$

Therefore

$$\binom{|A|}{2} + \binom{|B|}{2} \geq (1 - o(1)) \binom{f(N)}{2},$$

as claimed. This is incompatible with any upper bound of the form $(1 - c + o(1)) \binom{f(N)}{2}$ with fixed $c > 0$. □

REFERENCES

- [1] R. C. Bose and S. Chowla, *Theorems in the additive theory of numbers*, Comment. Math. Helv. **37** (1962/63), 141–147.
- [2] T. F. Bloom, *Erdős Problem #42*, <https://www.erdosproblems.com/42>, accessed 29 April 2026.
- [3] T. F. Bloom, *Erdős Problem #42 discussion thread*, including comments by Terence Tao, <https://www.erdosproblems.com/forum/thread/42>, accessed 29 April 2026.
- [4] T. F. Bloom, *Erdős Problem #43*, <https://www.erdosproblems.com/43>, accessed 29 April 2026.
- [5] T. F. Bloom, *Erdős Problem #43 discussion thread*, including comments by Kevin Barreto and Terence Tao, <https://www.erdosproblems.com/forum/thread/43>, accessed 29 April 2026.
- [6] P. Erdős and P. Turán, *On a problem of Sidon in additive number theory, and on some related problems*, J. London Math. Soc. **16** (1941), 212–215.